

UDDEEPTA DEKA

CONTACT INFORMATION

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RESEARCH INTERESTS

Gravitational wave (GW) astrophysics: gravitational lensing, waveform modeling, parameter estimation; machine learning: surrogate models; compact objects, alternative gravity theories.

EDUCATION

Ph.D. (Physics), ICTS - TIFR ¹, Bengaluru, Karnataka. India Aug 2019 – Dec 2025
Supervisor: Prof. Parameswaran Ajith
Thesis title: *Microlensing of gravitational waves as a new tool for astrophysics.*
M.Sc. Physics, University of Delhi, Delhi. India Aug 2016 – July 2018
B.Sc. Physics, Ramjas College, University of Delhi, Delhi. India July 2013 – July 2016

EMPLOYMENT

Assistant Researcher, Shanghai Astronomical Observatory - CAS. Mar 2026 – present
Research Associate, ICTS - TIFR. Aug 2025 – Dec 2025
Junior Research Fellow, University of Delhi. Aug 2018 – July 2019

AWARDS AND FELLOWSHIPS

- **DAE² PhD fellowship**: grant for pursuing PhD in DAE institutes. Aug 2019
- **CSIR-JRF³**: national-level grant for pursuing Ph.D. in India. Apr 2018
- **Delhi University Faculty of Science Meritorious Award 2016-17**: securing position among Top 10 students in M.Sc. Physics class in semesters 1& 2. Feb 2018
- **Indian Academy of Sciences Summer Research Fellowship**: national-level grant for working on a summer project at the National Centre for Radio Astrophysics - Tata Institute of Fundamental Research. Sep 2017
- **UGC-JRF⁴**: national-level grant for pursuing Ph.D. in India. Apr 2017
- **Delhi University Faculty of Science Meritorious Award 2013-16**: securing position among Top 10 students in B.Sc. Physics. Feb 2017
- **Ramjas college Academic Award - Gold Medal**: securing highest score in B.Sc.(honors) Physics, Ramjas College. Jan 2017

SELECTED SEMINARS, COLLOQUIA AND POSTERS

- INVITED TALKS
 - *Lensed gravitational-waves as astrophysical probes*, Department of Physics and Astrophysics, University of Delhi, Delhi, India. Feb 2026
 - *Astrophysics with microlensing of gravitational waves*, Indian Association for the Cultivation of Science (IACS), Kolkata, India. Aug 2025
 - *Astrophysics with microlensing of gravitational waves*, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India. July 2025
 - *Microlensing of gravitational waves: prospects and challenges*, Cosmology group seminar, Max Planck Institute for Gravitational Physics (AEI), Potsdam, Germany. Nov 2024
 - *Microlensing of gravitational waves: prospects and challenges*, Ghent University, Ghent, Belgium. Dec 2024

¹International Centre for Theoretical Sciences, Tata Institute of Fundamental Research

²Department of Atomic Energy, Government of India

³Council of Scientific and Industrial Research - Junior Research Fellowship

⁴University Grants Commission - Junior Research Fellowship

- *Workshop on Gravitational Wave data analysis*⁵, Delhi Technological University, Delhi, India. Feb 2023

- CONTRIBUTED TALKS

- *Impact of the lens galaxy on the search for gravitational-wave microlensing by compact objects*, *2nd New Horizons for Psi workshop*. ICTP-AP, Beijing, China. Aug 2026
- *Efficiency of current gravitational wave microlensing searches using isolated compact object lenses*, *CosPA & ACGRG*. Uni. Canterbury, New Zealand. July 2026
- *Microlensing of gravitational waves: prospects and challenges*, *ICTS in-house symposium*. ICTS, Bengaluru, India. Apr 2024
- *Microlensing of gravitational wave signals: effect of the galaxy lens*, *ICTS in-house symposium*. ICTS, Bengaluru, India. Apr 2023
- *Fast and accurate parameter estimation of gravitationally lensed gravitational-wave signals*, *32nd meeting of the Indian Association for General Relativity and Gravitation*. IISER, Kolkata, India. Dec 2022

- POSTER PRESENTATIONS

- *Is the isolated point mass a good model for gravitational-wave microlensing searches?*, *The Future of Gravitational-Wave Astronomy*, ICTS, Bengaluru, India. Oct 2025
- *Probing black hole hair with gravitational microlensing of gravitational waves*, *10th International Conference on Gravitation and Cosmology: New Horizons and Singularities in Gravity (ICGC 2023)*, IIT Guwahati, India. Dec 2023

TEACHING AND MENTORSHIP EXPERIENCE

- Teaching Assistant (TA) to Prof. Surhud More, *Cosmography using GW standard sirens*. Jul 2025
- TA to Prof. Dimitrios Psaltis, *The Kerr metric and its observational consequences*. Mar-Apr 2025
- TA to Prof. T. W. Baumgarte, *Numerical relativity: mathematical formulation*. Jul-Aug 2023
- Mentor at the *GW Open Data Workshop study hub* at ICTS, Bengaluru. May 2023
- Co-supervisor of MSc student Shivashree Shetty - project on *GW detection and parameter estimation*. Aug 2022 - Jun 2023
- TA to Prof. B. Mueller, *Stellar structure, evolution, collapse and supernovae*. May-Jun 2022
- TA to Prof. Vuc Mandic, *Stochastic astrophysical foreground from compact binary mergers*. Jul 2021
- TA to Prof. Loganayagam R., *ICTS graduate course - adv. Electrodynamics*. Jan-Jul 2021
- TA to Prof. R. Nityananda, *ICTS Summer course - Light and Beyond*. Jul-Aug 2020

PROFESSIONAL MEMBERSHIPS

- *Member*, *LIGO scientific collaboration*. Dec 2022 – July 2026
Role: Develop inference tools and detection pipelines for gravitational wave (GW) data. Organize workshops on GW data analysis. Part of the LIGO-India Education and Public Outreach program. Design science demonstrations.
- *Member*, *KAGRA collaboration*. July 2026 – present

SKILLS AND SPECIALIZATION

- **Technical:** *Computational statistics* | *Numerical methods* | *Signal processing* | *Data analysis* | *Scientific modeling* | *Machine learning*
- **Programming Languages:** EXPERIENCED: *Python*, *Mathematica* | FAMILIAR: *C++*, *C*.
- **Operating Systems:** EXPERIENCED: *Linux*, *MacOS* | FAMILIAR: *MS Windows*
- **Python Libraries:** *Numpy* | *Scipy* | *Scikit-learn* | *TensorFlow* | *Pandas* | *Matplotlib* | *LALInference* | *Astropy* | *PyCBC* | *BILBY* | *Lenstronomy*
- **Other software:** *LaTeX* | *Git* | *Docker*

⁵As part of LIGO-India Education and Public Outreach

PUBLICATIONS

Short-author publications:

- **U. Deka**, S. Chakraborty, S. J. Kapadia, M. A. Shaikh, P. Ajith, *Probing the charge of compact objects with gravitational microlensing of gravitational waves*, *Phys. Rev. D* **111**, 064028, [arXiv:2401.06553](#). 2025
- **U. Deka**, G. Prabhu, M. A. Shaikh, S. J. Kapadia, V. Varma, S. E. Field, *Surrogate modeling of gravitational waves microlensed by spherically symmetric potentials*, *Phys. Rev. D* **111**, 104042, [arXiv:2501.02974](#). 2025
- G. Prabhu, **U. Deka**, S. Chakraborty, S. J. Kapadia, *Probing the spin of compact objects with gravitational microlensing of gravitational waves*, [arXiv:2512.18707](#). 2025
- S. Konar, **U. Deka**, *Radio Pulsar Sub-Populations (I) : The Curious Case of Nulling Pulsars*, *J. Astrophys. Astron.* **40**, 42, [arXiv:1908.07681](#). 2019

Selected long-author publications (with direct contribution):

- LIGO Scientific, Virgo and KAGRA Collaborations (incl. **U. Deka**), *GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients*, *Astrophys. J. Lett.* **1004**, L21, [arXiv:2508.18081](#). 2026
- LIGO Scientific, Virgo and KAGRA Collaborations (incl. **U. Deka**), *GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*, *Astrophys. J. Lett.* **1004**, L22, [arXiv:2508.18082](#). 2026
- LIGO Scientific, KAGRA and Virgo Collaborations (incl. **U. Deka**), *GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog*, *Astrophys. J. Lett.* **995**, L18, [arXiv:2508.18080](#). 2025
- LIGO Scientific, Virgo and KAGRA Collaborations (incl. **U. Deka**), *GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures*, [arXiv:2512.16347](#). 2025